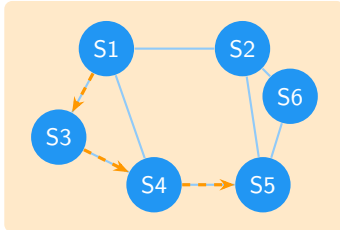


Input

	C1	C2	C3	C4	...
S1	1	0	1	0	...
S2	0	1	1	0	...
S3	1	1	0	1	...
S4	0	0	1	1	...
S5	1	0	1	0	...
S6	0	1	0	1	...

Student $\times$ Course Matrix

Student Graph



Nodes: Students, Edges: Shared Courses

Node2Vec

Random Walk Sequences:

S1 $\rightarrow$ S3 $\rightarrow$ S4 $\rightarrow$ S5

S2 $\rightarrow$ S5 $\rightarrow$ S6 $\rightarrow$ S5

S3 $\rightarrow$ S1 $\rightarrow$ S2 $\rightarrow$ S6

Word2Vec (Skip-Gram)

Learn Node Embeddings

Key Results

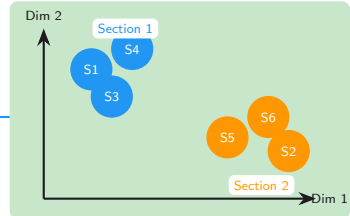
Silhouette Score

+35% vs. PCA

Section sizes: Comparable

Impact: Poor $\rightarrow$ Good clustering

2D Embeddings



Separable Clusters